ALICE L WALTON SCHOOL OF MEDICINE

Physiology of Heart Failure

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Adapted from: Physiology of Heart Failure

Listen to this Brick

0:00 / 35:25 1x

Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Describe the causes and types of cardiac hypertrophy.
2. Explain how the pattern of sarcomeres creates concentric and eccentric cardiac hypertrophy and how each leads to heart failure.
3. Outline the causes and consequences of ischemic heart injury.
4. Describe the ow heart failure causes activation of the renin-angiotensin-aldosterone system (RAAS) and sympathetic nervous system.
5. Explain the changes that occur to pressure-volume loops in patients with acute heart failure.

CASE CONNECTION

DR is a 56-year-old male with longstanding hypertension who presents to the emergency department with difficulty breathing and swelling in his legs. "I can barely walk to my truck anymore because my legs are so heavy and swollen, and I can't catch my breath." He tells you that he stopped taking his water pills because he lost his job and can't afford his medications. DR's vital signs are heart rate of 96/min, blood pressure 152/94 mm Hg, and respiratory rate 22/min. On exam, he has 2+ pitting edema in his lower extremities and bilateral rales.

What treatment should DR receive? Consider your answer as you read, and we'll revisit DR at the end of the brick.

GO TO CONCLUSION ↓

What Is Heart Failure (LO1)?

Heart failure (HF) occurs when the heart can't pump enough blood to meet the needs of the body. There are really 2 questions that highlight the physiologic approach to heart failure:

- What happens w d?
- What happens when ventricular filling is impaired?

The myocytes in the heart wall (myocardium) respond to injury in several ways, with two mechanisms accounting for almost all heart failure cases: hypertrophy and remodeling after ischemic injury.

Myocardial Hypertrophy – Concentric and Eccentric (LO2)?

The change in myocyte size is called **hypertrophy**. In heart failure, there is an increase in the number of **sarcomeres**, the basic contractile unit for cardiac myocytes (Figure 1). While some hyperplasia (increase in cell number) may occur in heart disease, hypertrophy is the major mechanism for cardiac enlargement. There are two types of cardiac hypertrophy based on how sarcomeres are added: **concentric** (parallel arrangement) and **eccentric** (series arrangement).

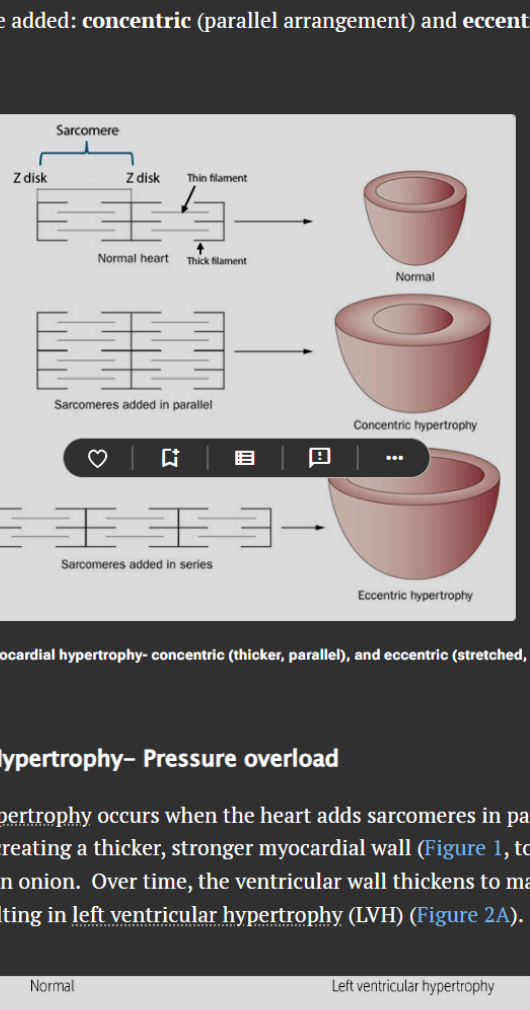
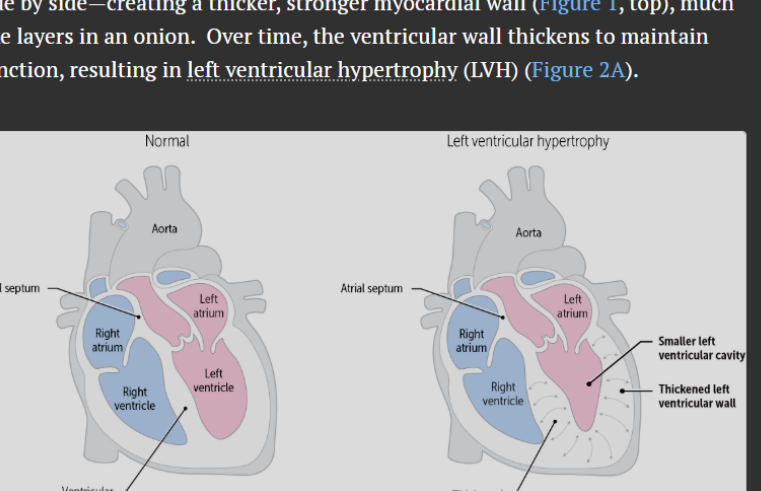


Figure 1. Myocardial hypertrophy- concentric (thicker, parallel), and eccentric (stretched, series)

Concentric Hypertrophy– Pressure overload

Concentric hypertrophy occurs when the heart adds sarcomeres in parallel—side by side—creating a thicker, stronger myocardial wall (Figure 1, top), much like layers in an onion. Over time, the ventricular wall thickens to maintain function, resulting in left ventricular hypertrophy (LVH) (Figure 2A).



Causes of Concentric Hypertrophy

Chronic conditions that raise afterload:

- Chronic, uncontrolled hypertension (most common)
- Aortic stenosis
- Hypertrophic cardiomyopathy (rare genetic disorder)

Eccentric Hypertrophy– Volume overload

In contrast, eccentric hypertrophy adds sarcomeres in **series** (end-to-end), leading to a thinner, dilated ventricular wall (Figure 1, bottom). This process is more commonly termed "ventricular dilation," and the heart myocardium is stretched thinner and appears like a balloon (Figure 2B). Eccentric hypertrophy can sometimes be seen in normal situations like in highly conditioned athletes, but it is often pathological when there is marked volume increase in the ventricles.

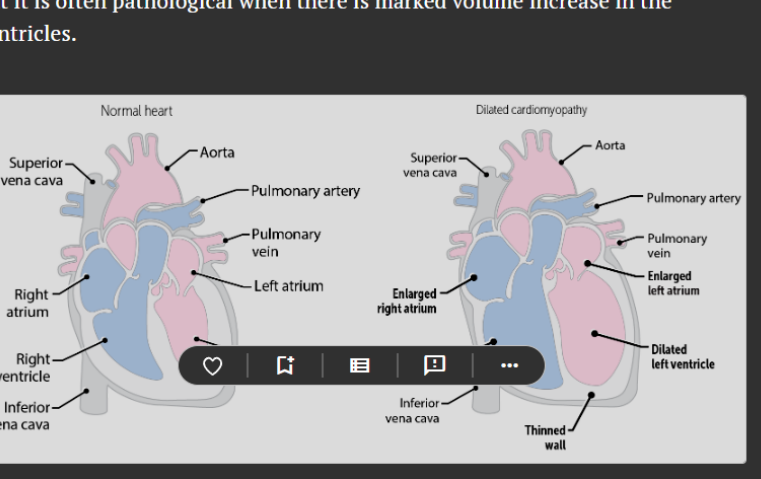


Figure 2B. Eccentric hypertrophy; dilated cardiomyopathy

How does sarcomere arrangement differ between concentric and eccentric hypertrophy?

Causes of Eccentric Hypertrophy

Eccentric hypertrophy occurs either due to volume overload or myocyte injury:

Volume overload

- Mitral or aortic valve regurgitation - the ventricle must expel both normal blood from the lungs plus extra blood that regurgitates back through incompetent "leak
- End-stage kidney by the kidney is dysfunctional.

Myocyte injury:

- Ischemia (most common)
- Viral infections
- Myocarditis
- Alcohol
- Certain drugs (e.g., doxorubicin)

CLINICAL REASONING

While pathologically almost all heart failure involves some form of cardiac hypertrophy, the term "left ventricular hypertrophy (LVH)" is clinically reserved for concentric hypertrophy. Eccentric hypertrophy is typically called "ventricular dilation" or "dilated cardiomyopathy."

The Price of Adaptation: When Hypertrophy Causes Ischemia

Cardiac hypertrophy initially helps the heart adapt to stress, but over time these changes become detrimental. Hypertrophic hearts develop low cardiac perfusion through two mechanisms:

- **Decreased capillary density:** New blood vessels do not increase proportionally with the hypertrophied muscle mass
- **Arterial compression:** The extra cardiac mass compresses coronary arteries, reducing

This creates a vicious cycle: hypertrophy leads to tissue hypoxia and ischemia, which causes further myocyte injury and dysfunction. This is why hypertrophy itself can eventually lead to heart failure, even without additional insults like myocardial infarction. Additionally, cardiac hypertrophy increases oxygen demand due to increased muscle mass and elevated wall stress, which can also contribute to ischemia.

How does cardiac hypertrophy lead to myocyte ischemia?

LaPlace law– wall tension and wall stress

The Law of Laplace explains how the heart adapts to pressure and volume overload. It's essential to distinguish between two related concepts:

Wall Tension (T) is the force on the ventricular wall (pressure × radius), but the clinically importa which represents the tension that is distributed across the thickness of the heart wall. Wall stress is the actual load on each cardiac muscle fiber; it's a **tensile stress within** the muscle fibers that must counteract the internal pressure. It is more physiologically relevant because it determines myocardial oxygen demand and cellular strain. By increasing wall thickness (h), the heart can reduce this wall stress and maintain function despite high pressure or volume.

$$\sigma = (P \times r)/2h$$

Concentric hypertrophy (thickening): With pressure overload, due to increased afterload, wall stress (σ) or tension (T) rises due to increased pressure (↑P) (Figure 3, top center). The heart thickens in response, adding sarcomeres in parallel (like onions, they have layers). This helps normalize wall stress (σ = T/2h) despite high pressure (↑P), maintaining systolic function. *note: *You do not need to memorize this equation, if the equation helps- then great. But you need to be aware of the concept behind it: factors that contribute to wall stress. Indeed, these are the factors that you learn that are crucial in the different types of heart failure. Also that wall stress increases the demand on the heart**.

- **Disadvantage:** The thickened, stiff, less compliant wall impairs filling, leading to **diastolic dysfunction**—the hallmark of **heart failure with preserved ejection fraction (HFpEF).**

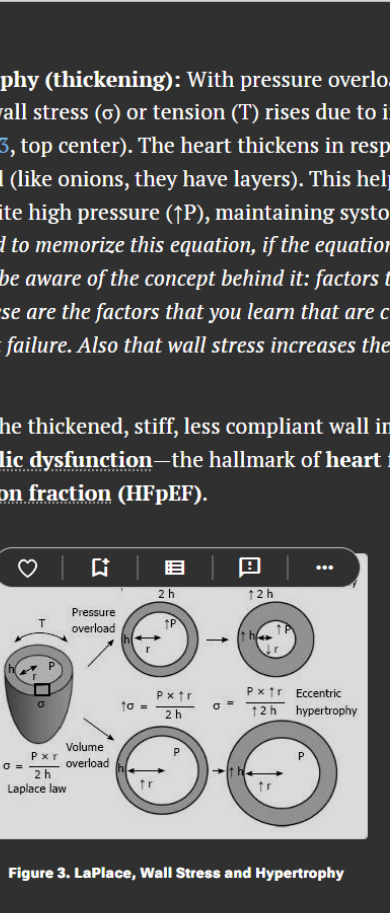

T
Pressure overload
h
P
r
σ
$\sigma = \frac{P \times r}{2h}$
Laplace law
2 h
↑P
$\uparrow\sigma = \frac{P \times \uparrow r}{2h}$
↑2 h
$\sigma = \frac{P \times \uparrow r}{\uparrow 2h}$
Eccentric hypertrophy
Volume overload
↑r
↑h
P

Figure 3. LaPlace, Wall Stress and Hypertrophy

Eccentric hypertrophy (dilation): With volume overload (e.g., regurgitation), the increased volume (↑EDV) enlarges the radius (↑r), which then increases wall stress (↑σ) (Figure 3 , bottom center). The ventricles respond by dilating further (↑r), adding sarcomeres in series, with some accompanying wall thickening (Figure 3, bottom right).

- **Disadvantage:** Over time, dilation outpaces wall thickening, the wall becomes thinned, causing wall stress (↑σ) to rise and contractility to decline. This leads to **systolic dysfunction**—the hallmark of **heart failure with reduced ejection fraction (HFrEF).**

What is the most common cause of concentric cardiac hypertrophy?

Frank Starling Law of the heart

The Frank–Starling curve illustrates how stroke volume changes with end-diastolic volume (EDV) (Figure 4). In a normal ventricle, increasing EDV stretches sarcomeres to a more optimal length, boosting contraction and cardiac output.

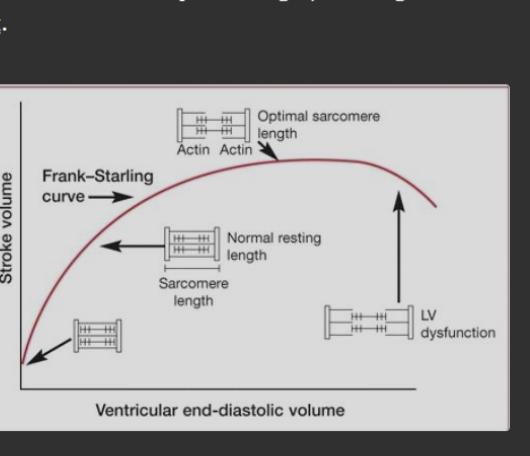


Figure 4. Frank Starling Curve of the heart- stretch affects the force of contraction

- **In a dilated ventricle** (eccentric hypertrophy), sarcomeres are already overstretched. Adding more volume pushes the heart onto the descending limb of the curve, where stroke volume falls despite high filling—typical of HFrEF. Adding sarcomeres in series does have some benefits early on: It allows the myocytes to function at a more optimum sarcomere length rather than an overstretched length. With little increase in wall thickness, the increased diameter is associated with increased wall stress, further damaging and weakening the dilated heart.
- **In stiff, hypertrophied ventricles** (concentric hypertrophy), the heart cannot efficiently expand to take in more blood, so EDV is reduced. These hearts operate at the low end of the ascending limb. Stroke volume is reduced, but contractility is preserved—typical of HFpEF.

What Is the Pathophysiology of Ischemic Injury (LO3)?

Ischemic heart disease (low oxygen delivery to the heart) is the most common cause of heart failure. Necrosis of myocytes and heart failure can occur not just after myocardial infarctions (heart attacks) but also after chronic ischemic injury without overt infarction, such as when coronary artery disease leads to ischemia. Heterogeneous **cardiac remodeling** then occurs to replace the dead myocytes. The tissue deposited during this remodeling is disorganized, with fibrosis replacing muscle. Both diastolic dysfunction (impaired filling) and systolic dysfunction (impaired contraction) can result.

Reduced End-Diastolic Volume

In **diastolic dysfunc** pedes filling and ventricular relaxation—the heart is stiff and noncompliant. These effects impair the ventricular filling that occurs during diastole, which reduces the end-diastolic volume (EDV) and lowers stroke volume (SV) and cardiac output.

Myocardial ischemia can also impair ventricular relaxation. Myocardial relaxation requires ATP because calcium must be actively pumped out of the cytosol during diastole by the SERCA. The SERCA pump transports Ca^{2+} back into the sarcoplasmic reticulum against its concentration gradient, ending contraction, and so relaxation is an energy-dependent process.

Decreased Contractility

In **systolic dysfunction**, the disordered cellular architecture impairs sarcomere coordination. This reduces the overall quality of ventricular contractions during systole. As a result, blood does not move forward efficiently, meaning some of it pools in the ventricular chamber.

Extra ventricular pressure from the pooling blood further stimulates the heterogeneous cardiac remodeling process. Together, these factors worsen myocardial function, leading to decreased contractility (the innate ability of the myocardium to pump blood out of the heart), decreased stroke volume, and decreased cardiac output.

How Does the Body Respond to the Failing Heart (LO4)?

When heart failure develops, other organs do not receive adequate blood flow and respond in ways that lead to classic heart failure symptoms. To understand this response, we use the concept of **effective circulating volume**—the circulating blood tha ot measurable but is a physiologic response determined by cardiac output, blood pressure, and blood volume.

When heart failure causes decreased cardiac output, the body compensates by increasing blood pressure and blood volume to raise effective circulating volume back toward normal. However, this compensation causes troublesome symptoms like edema and dyspnea.

Renin–Angiotensin–Aldosterone System and Sympathetic Nervous System (LO4)

The body responds to heart failure and decreased effective circulating volume through two major pathways (Figure 5):

- **Renin-Angiotensin-Aldosterone System (RAAS):** Increases intravascular pressure through angiotensin-mediated vasoconstriction and aldosterone-mediated fluid retention
- **Sympathetic Nervous System (SNS):** Increases heart rate, contractility, and arteriole vasoconstriction

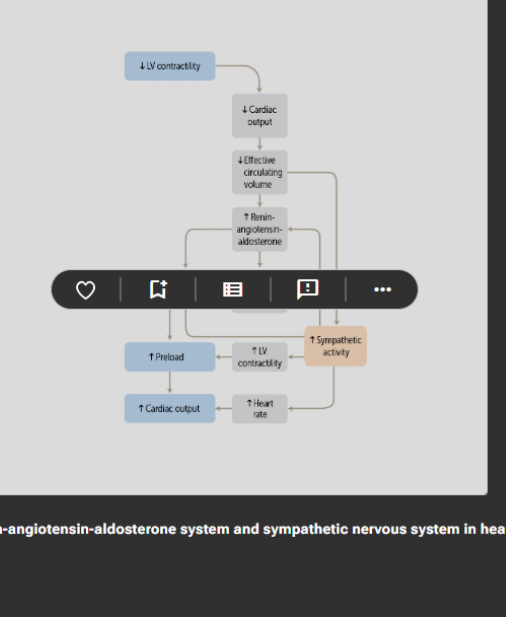


Figure 5 Renin-angiotensin-aldosterone system and sympathetic nervous system in heart failure

Renin-Angiotensin-Aldosterone System. A drop in cardiac contractility causes decreased cardiac output, which decreases effective circulating volume and activates RAAS via reduced renal blood flow (Figure 5). This increases renin secretion in the kidney, triggering an increase in angiotensin II (AT-II).

Angiotensin II has three major effects:

- Constricts peripheral blood vessels (raising arterial and venous pressure)
- Stimulates the posterior pituitary to release antidiuretic hormone (ADH, or vasopressin), increasing renal water reabsorption
- Stimulates the adrenal cortex to secrete aldosterone. Aldosterone raises blood volume by increasing renal reabsorption of sodium and water, which also increases blood pressure.

Result: The increased blood volume and venous pressure raise venous return to the heart and increase cardiac preload. This maximizes cardiac output via the Frank-Starling mechanism, even with poor contractility.

Sympathetic Nervous System. Decreased cardiac output leads to low blood pressure, stimulating carotid and aortic baroreceptors and activating the SNS. Additionally, as the h chambers, causing muscle stretch that also activates the SNS. SNS enhances cardiac output through:

- Increased heart rate (via β_1 receptors)
- Increased cardiac contractility (via β_1 receptors-although this may not work, since a drop in contractility is why the heart initially failed)
- Increased systemic vascular resistance through vasoconstriction, which increases venous tone and venous return

Summary: SNS, RAAS, and kidneys work together to increase heart rate and preload, improving failing cardiac output. We should note that while initially compensatory, chronic activation of the RAAS and SNS leads to pathological remodeling, fibrosis, and progressive worsening of heart failure.

How does chronic activation of RAAS contribute to worsening heart failure?

The Paradox of Co ic Harm

The body's responses to heart failure illustrate a critical principle in physiology: mechanisms that save lives acutely become destructive chronically.

Acute Phase (Hours to Days): Life-Saving. RAAS and SNS activation maintain blood pressure and organ perfusion. Increased preload maximizes stroke volume via the Frank-Starling mechanism, while increased heart rate maintains cardiac output despite reduced stroke volume. These responses can be life-saving in acute heart failure.

Chronic Phase (Weeks to Years): Maladaptive. Persistent vasoconstriction increases afterload (and blood pressure), making the failing heart work harder against higher resistance. Fluid retention due to RAAs activation, and heart dysfunction, leads to pulmonary edema (causing dyspnea) and peripheral edema (causing leg swelling)—classic heart failure symptoms. Chronic RAAS activation promotes myocardial fibrosis, collagen deposition, and pathological remodeling. Chronic SNS activation causes direct cardiomyocyte damage and apoptosis, LV dilation and systolic dysfunction, and further decreases in cardiac output. These maladaptive changes create a vicious cycle of worsening heart failure.

CLINICAL CORRELATION

This explains why heart failure medications work—ACE inhibitors, beta-blockers, mineralocorticoid antagonists (which block aldosterone and reduce sodium and water retention), and loop diuretics interrupt these chronic maladaptive processes while potentially sacrificing some acute compensatory benefit.

Loop diuretics inhi porter (NKCC) and increase sodium and and relieving edema.

Mineralocorticoid antagonists block aldosterone and reduce sodium and water retention.

ACE inhibitors block RAAS, reducing afterload and preventing fibrosis.

Beta-blockers block SNS, preventing chronic cardiomyocyte damage.

How does RAAS activation worsen heart failure?

How Can We Explain Heart Failure Using the Pressure–Volume Loop (LO5)?

Heart failure may result from impaired contraction (systolic dysfunction) or impaired filling (diastolic dysfunction). In both cases, stroke volume (SV) is reduced because less ardiac output (CO) = SV × HR, output may be maintained early in heart failure if the heart rate rises to compensate.

Clinically, these patterns correspond to two major categories that map directly onto pressure-volume loop changes:

- **Heart failure with reduced ejection fraction (HFrEF):** reduced contractility and lower stroke volume.
- **Heart failure with preserved ejection fraction (HFpEF):** impaired filling due to stiffness, reducing stroke volume despite normal contractility.

We can understand these complex concepts by visually representing them with pressure-volume loops.

Systolic Heart Failure – Volume Overload

In systolic heart failure (diastolic HF with eccentric hypertrophy and a dilated ventricle) reduced myocardial contractility fundamentally alters the pressure-volume loop (Figure 6, orange loop). The **end-systolic pressure-volume relationship (ESPVR)** becomes flatter, reflecting diminished pressure generation for any given ventricular volume. This reduced contractility causes the heart to pump out less blood; the end-systolic volume (ESV) increases markedly while end-diastolic volume (EDV) increases slightly. The net result is decreased stroke volume (SV = EDV - ESV), primarily due to elevated ESV. Since ejection fraction (EF = SV/EDV) is reduced, this condition is termed heart failure with reduced ejection fraction (HFrEF). Normal ejection fraction is approximately 60%.

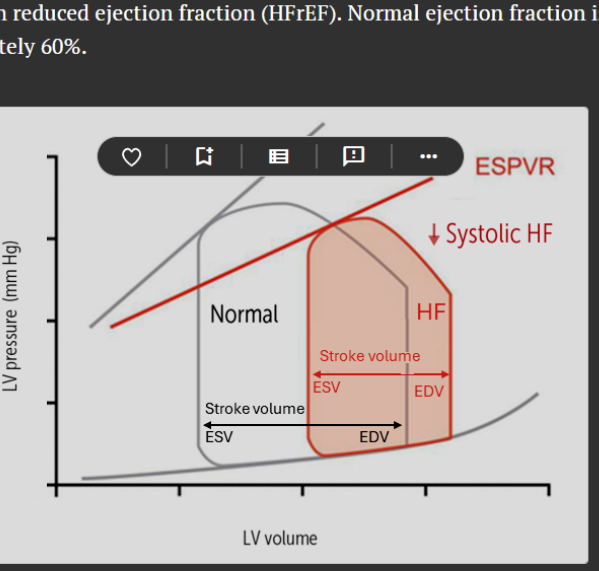


Figure 6. Systolic heart failure

The Preload Paradox. Intuitively, if we have low stroke volume, an increase in preload (EDV) would increase the force of contraction via the Frank-Starling mechanism. However, the stiffer ventricle results in the end-diastolic pressure-volume relationship (EDPVR) shifting to curve sharply upward, leaving minimal capacity for additional fluid. Administering more volume to patients with HFrEF may precipitate pulmonary edema due to elevated hydrostatic pressures. This illustrates the fundamental challenge in heart failure management: compensatory mechanisms often carry significant adverse consequences.

In which type of heart failure is preload typically **increased** and why?

Diastolic Heart Failure – Pressure overload

What about diastolic heart failure, where concentric hypertrophy (a thicker and stiffer heart ventricle) limits ventricular filling? This pressure-volume loop is shown in Figure 7.

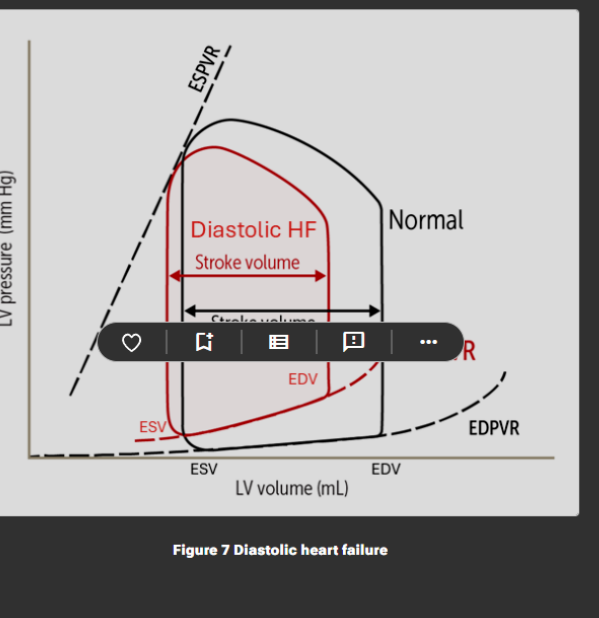


Figure 7 Diastolic heart failure

Notice that there is a decrease in the EDV, which occurs because hypertrophy limits diastolic filling of the LV. The contractility stays the same, so the slope of the contractility curve (ESPVR) is unchanged. In a sense, this new loop looks similar to the loop for reduced preload, and the stroke volume is similarly reduced.

Let's look at the volumes in the heart. The heart fills with less blood, so the end-diastolic volume (EDV) decreases. If you compare the stroke volume (EDV-ESV) for the two curves, you will see that the stroke volume has dropped, largely due to the fall in EDV due to reduced filling of the stiffened heart. This decrease in SV is why the body tries to increase the heart rate, so that even with the drop in SV, the cardiac output (SV x HR) can be maintained. The amount of blood ejected by the heart can be measured as ejection fraction, which is the stroke volume/ EDV. In this case BOTH the SV and EDV are reduced, and the EF is preserved. Thus it is termed heart failure with preserved ejection fraction (**HFpEF**). A normal ejection fraction is about 60%.

Please note: the LV diastolic pressure is now higher at any given volume. Why? The hypertrophy and stiff ventricle mean that any amount of added volume increases the LV pressure shifts upward. The end result is that SV the lower EDV, which is why cardiac output falls in diastolic dysfunction, even though contractility is normal.

CLINICAL REASONING

Pressure–volume loops and the Frank–Starling curve are two views of the same physiology.

In **Systolic HF or HFrEF**, the PV loop shows increased EDV and reduced stroke volume. This corresponds to a dilated ventricle sitting on the descending limb of Starling.

In **Diastolic HF or HFpEF**, the PV loop shows reduced EDV but preserved contractility. This corresponds to a stiff ventricle stuck on the low part of the ascending limb.

How do PV loops, Laplace's law, and Starling's law connect?
They all describe how the heart adapts — and fails — under pressure and volume load.

Why is ejection fraction preserved in HFpEF even though patients have reduced exe eart failure?

How do PV loops, Laplace's law, and Starling's law connect?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to DR, what treatment should he receive?

His chest x-ray shows that DR has heart failure secondary to concentric left ventricular hypertrophy from his longstanding hypertension. He is admitted to the hospital and treated with diuretics. When you see him the next day, DR smiles and says, "I'm peeing a t ood my legs look!" You stress the impor ct him with a social worker to help him obtain them.

Summary

- Heart failure occurs when the heart cannot pump enough blood to meet the body's needs, due to either systolic or diastolic dysfunction.
- Ischemia occurs when myocardial oxygen supply is insufficient to meet demand. In conditions such as coronary artery disease or hypertrophy, this imbalance leads to maladaptive remodeling of cardiac myocytes.
- Compensatory changes to heart failure occur via 1. either concentric (thickening of walls) due to chronic hypertension, or 2. eccentric (dilation of walls) due to high volume.
- The body also responds to the failing heart by activating the renin-angiotensin-aldosterone system and sympathetic nervous system, leading to increased preload and heart rate.
- These compensatory actions in the acute phase may be life saving but then chronically, over time they become maladaptive, contributing to further heart failure.
- Changes in heart failure are represented by pressure-volume loops, with systolic failure reducing contractility and stroke volume (less ejection increases preload), and diastolic failure reducing preload and stroke volume.

Review Questions

1. A 50-year-old male ilated or eccentric cardiomyopathy (elongated sarcomeres arranged in series), characterized by severe systolic heart failure. Which of the following is likely to be most increased in this patient?

   A. ○ Contractility
   B. ○ Ejection fraction
   C. ● End-diastolic volume
   D. ○ Stroke volume

   Hide Explanation

   The correct answer is end-diastolic volume (C). Because of the decrease in contractility (A) in systolic heart failure, the heart is not able to pump blood out as effectively. This means that more blood will remain in the heart after systole, and blood will be added during diastole. The end-diastolic volume is, therefore, increased. The ejection fraction (B), or percentage of the volume of blood pushed out of the heart during systole, is decreased in systolic heart failure. Stroke volume (D) decreases because it is the difference between end-diastolic volume (EDV) and end-systolic volume (ESV). The EDV increases a little, but the ESV increases even more. The difference between EDV and ESV decreases.

2. Which of the following is the most common cause of concentric left ventricular hypertrophy (thicker llel)?

   A. ○ Arrhythmias
   B. ○ Exercise
   C. ● Hypertension
   D. ○ Myocardial infarction
   E. ○ Pregnancy

   Hide Explanation

   The correct answer is hypertension (C). Hypertension is the leading cause of left ventricular hypertrophy (LVH). Arrhythmias (A) do not in themselves cause LVH. Although exercise (B), remodeling after myocardial infarction (D), and pregnancy (E) can all cause LVH, they are not as common as hypertension.

3. On a pressure-volume loop, what does the end systolic pressure-volume relationship curve represent?

   A. ○ Afterload
   B. ● Contractility
   C. ○ Preload
   D. ○ Stiffness
   E. ○ Stroke volume

   Hide Explanation

   The correct answer is contractility (B). The end-systolic pressure-volume relationship (ESPVR) is the upper boundary line of the pressure-volume loop. It represents the pressure that can be generated for a given volume in the ventricle. In chronic heart failure, the slope of this line is decreased, because at a given volume the heart generates less force. Afterload (A) is a measure of the pressure the ventricle is pumping against and is measured as the point of intersection of the pressure-volume loop with the ESPVR curve. Preload (C) is the amount of volume in the ventricle just before systole, and it is shown by the intersection of the pressure-volume loop with the end diastolic pressure volume relationship (EDPVR) curve. Ventricular stiffness (D) is a measure of the ventricle's ability to relax or fill with blood and is represented by the EDPVR curve. Stroke volume (E) is the amount of blood ejected from the ventricle with each contraction. It is measured as end-diastolic volume-end-systolic volume (EDV-ESV) and is also measured as the width of the pressure-volume loop.

4. Which of the following is the most likely consequence of chronic renin-angiotensin-aldosterone system (RAAS) activation in heart failure?

   A. ○ Decreased afterload
   B. ○ Decreased arteriole vasoconstriction
   C. ○ Decreased preload
   D. ● Increased afterload
   E. ○ Increased renal sodium excretion

   Hide Explanation

   The correct answer is increased afterload (D). Renin-angiotensin-aldosterone system (RAAS) activation in heart failure has two overarching effects: increased preload and increased afterload, not decreased afterload (A). This is accomplished by increased, not decreased, arteriole vasoconstriction (B), which is mediated by angiotensin. RAAS activation leads to increased, not decreased, preload (C). It also causes increased sodium absorption via aldosterone, not increased renal sodium excretion (E). Although these effects are initially adaptive to try and maintain cardiac output, in the long run, they become harmful because they cause increased congestion and make the failing heart pump against more afterload.

5. A 62-year-old man with long-standing coronary artery disease presents with gradually worsening exertional dyspnea over several months. He has no history of an acute myocardial infarction. Physical examination reveals bibasilar crackles and an S4 heart sound. Echocardiography shows reduced ventricular compliance with impaired relaxation and mildly reduced stroke volume. Which of the following pathophysiologic processes most *directly* explains this patient's impaired ventricular relaxation?

   A. ● ATP-depe c reticulum is impaired
   B. ○ Decreased capillary density leads to reduced coronary perfusion during diastole
   C. ○ Increased end-diastolic volume stretches sarcomeres beyond optimal length
   D. ○ Parallel addition of sarcomeres increases ventricular wall thickness

   Hide Explanation

   The correct answer is (A) The S4 heart sound indicates diastolic HF (impaired filling of the heart), and MI is the most common cause of concentric hypertrophy. Myocardial relaxation during diastole requires ATP for calcium reuptake via SERCA; ischemia and ischemic damage reduce ATP availability, directly impairing ventricle relaxation and causing diastolic dysfunction. (B) Reduced capillary density contributes to ischemia but does not *directly* explain impaired relaxation at the cellular level. (C) Sarcomere overstretching occurs in systolic dysfunction with dilation (eccentric hypertrophy), not impaired relaxation. (D) Parallel sarcomere addition is characteristic of concentric hypertrophy from pressure overload, not ischemic injury.

How would you rate this Brick?

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Review Learning Objectives

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☑ ~~Describe the causes and types of cardiac hypertrophy.~~
- 02 ☑ ~~Explain how the pattern of sarcomeres creates concentric and eccentric cardiac hypertrophy and how each leads to heart failure.~~
- 03 ☑ ~~Outline the causes and consequences of ischemic heart injury.~~
- 04 ☑ ~~Describe the body's responses to heart failure and how heart failure causes activation of the renin-angiotensin-aldosterone system (RAAS) and sympathetic nervous system.~~
- 05 ☑ ~~Explain the changes that occur to pressure-volume loops in patients with acute heart failure.~~

Objective Confidence: 5/5